

Dinesh

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PROFESSIONAL SUMMARY

Aspiring AI Engineer and Data Scientist with expertise in Machine Learning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and cloud-native AI deployment. Experienced in developing end-to-end AI systems using LangGraph, FastAPI, Docker, AWS, Supabase, Qdrant Cloud, and GitHub Actions. Built production-ready applications involving hybrid retrieval, Text-to-SQL, model fine-tuning, and MLOps automation. Strong statistical background with proven ability to design scalable, secure, and reliable AI solutions from experimentation to production deployment.

EDUCATION

Kurukshetra University

Aug 2024 – May 2026

M.Sc. in Statistics — Coursework: Time-Series Forecasting, Probability, Statistical Inference

DCRUST, Murthal

Aug 2020 – Jun 2023

B.Sc. in Mathematics — Coursework: Probability, DBMS, Linear Algebra, Calculus

TECHNICAL SKILLS

Languages: Python, SQL

GenAI/LLMs: RAG, LangChain, LangGraph, OpenAI API, Hugging Face, Fine-Tuning, Text-to-SQL, HyDE, CRAG

ML/DL: Scikit-Learn, XGBoost, TensorFlow, PyTorch

Backend: FastAPI, Flask, REST APIs, Pydantic

Databases: Supabase PostgreSQL, Qdrant Cloud, Upstash Redis

Cloud/DevOps: AWS (ECS, EC2, ECR, App Runner), Docker, GitHub Actions, CI/CD, Linux

MLOps: W&B, Weave, Model Deployment, Monitoring

PROJECTS

Enterprise RAG Platform for K8s IT Operations

GitHub

- Engineered a multi-agent RAG system using **LangGraph** to route complex queries between SQL and hybrid-retrieval workflows.
- Optimized retrieval precision via **Hybrid Search (Dense + Sparse)**, **HyDE**, and **Cross-Encoder Re-ranking**.
- Implemented a 9-layer security architecture, featuring **Human-in-the-Loop (HITL)** approval and prompt-injection defenses.
- Architected a multi-tier caching system using **Upstash Redis** to minimize latency in real-time IT operations.

Loan Approval Prediction System

GitHub

- Built a supervised machine learning solution to predict loan approval outcomes from customer demographic, income, employment, and credit history data.
- Conducted comprehensive data cleaning, feature engineering, and exploratory data analysis to identify key factors influencing loan approval decisions.
- Benchmarked multiple classification models and optimized performance using cross-validation and evaluation metrics such as Precision, Recall, F1-Score, and ROC-AUC.
- Developed a reusable prediction pipeline for real-time inference and decision support in lending applications.
- Generated actionable insights from model outputs to support risk assessment and improve loan approval decision-making processes.

Beverage Quality Prediction Platform

GitHub

- Built an end-to-end ML pipeline with automated data validation, transformation, and scalable model inference.
- Developed containerized **Flask** APIs to support modular deployment and system reliability.
- Orchestrated an automated MLOps lifecycle on **AWS EC2** using **Docker** and **GitHub Actions**.

INTERNSHIP EXPERIENCE

Data Analyst Intern

iMark Infotech Pvt Ltd

Data Analytics

Jun 2025 – Jul 2025

- Leveraged **Python (Pandas, NumPy)** and **SQL** to analyze complex datasets, surfacing actionable business insights to drive strategic decision-making.
- Engineered data cleaning, preprocessing, and **Exploratory Data Analysis (EDA)** pipelines, significantly improving data quality for downstream ML model training.
- Developed highly optimized **SQL queries** for large-scale data extraction and transformation, streamlining reporting across multiple cross-functional business domains.